Numerical Integration and LTI System Stability

Problem 1.2.19 (EE 2007)

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Problem Statement

The differential equation $\frac{dx}{dt} = \frac{1-x}{\tau}$ is discretised using Euler's numerical integration method with a time step $\Delta T > 0$.
What is the maximum permissible value of ΔT to ensure stability of the solution of the corresponding discrete time equation?

a) 1 b) $\frac{\tau}{2}$ c) τ d) 2τ

($\frac{dx}{dy}$ instead of $\frac{dx}{dt}$).

Euler's Forward Discretization

We begin with the fundamental definition of the forward difference approximation for the derivative:

$$\frac{dx}{dt} \approx \frac{x[n+1] - x[n]}{\Delta T} \tag{1}$$

Substituting this into our given differential equation:

$$\frac{x[n+1] - x[n]}{\Delta T} = \frac{1 - x[n]}{\tau} \tag{2}$$

Rearranging to form a linear first-order difference equation (a discrete-time LTI system):

$$x[n+1] = x[n] + \frac{\Delta T}{\tau}(1 - x[n]) \tag{3}$$

$$x[n+1] = \left(1 - \frac{\Delta T}{\tau}\right) x[n] + \frac{\Delta T}{\tau} \tag{4}$$

Z-Transform and System Pole

To rigorously analyze stability, we take the Z-transform of the homogeneous part of the difference equation, $x[n+1] - \left(1 - \frac{\Delta T}{\tau}\right) x[n] = 0$:

$$zX(z) - \left(1 - \frac{\Delta T}{\tau}\right) X(z) = 0 \tag{5}$$

The characteristic equation dictates the dynamics of the system:

$$z - \left(1 - \frac{\Delta T}{\tau}\right) = 0 \tag{6}$$

This yields a single system pole at:

$$p = 1 - \frac{\Delta T}{\tau} \tag{7}$$

The Unit Circle Stability Criterion

For any discrete-time LTI system to be asymptotically stable, its poles must lie strictly inside the unit circle on the Z-plane.

$$|p| < 1 \tag{8}$$

Substituting our pole location into the stability criterion:

$$\left|1 - \frac{\Delta T}{\tau}\right| < 1 \tag{9}$$

Expanding the absolute value inequality:

$$-1 < 1 - \frac{\Delta T}{\tau} < 1 \tag{10}$$

Solving for Maximum Time Step

Subtract 1 from all parts of the inequality:

$$-2 < -\frac{\Delta T}{\tau} < 0 \tag{11}$$

Multiply by $-\tau$ (Note: multiplying by a negative flips the inequality signs, assuming $\tau > 0$ for a physical system):

$$2\tau > \Delta T > 0 \tag{12}$$

$$0 < \Delta T < 2\tau \tag{13}$$

The strict limit for asymptotic stability is 2τ. If ΔT reaches or exceeds this, the discrete sequence diverges.

Correct Option: (d) 2τ

Python Computational Verification

To verify this boundary, we simulate the difference equation natively in Python across three distinct stability regimes.

```
import numpy as np
import matplotlib.pyplot as plt

def simulate_euler(tau, dt, steps, x0=0):
    x = np.zeros(steps)
    x[0] = x0
    for n in range(steps - 1):
        x[n+1] = (1 - dt/tau)*x[n] + dt/tau
    return x

tau = 1.0   # Set arbitrary time constant
steps = 25
n_axis = np.arange(steps)

# 1. Perfectly Stable (Pole at 0)
x_stable = simulate_euler(tau, dt=1.0, steps=steps)
# 2. Marginally Stable / Ringing (Pole at -0.9)
x_ringing = simulate_euler(tau, dt=1.9, steps=steps)
# 3. Unstable / Divergent (Pole at -1.1)
x_unstable = simulate_euler(tau, dt=2.1, steps=steps)

plt.figure(figsize=(8, 4.5))
plt.plot(n_axis, x_stable, 'b-o', label=r'$\Delta T = \tau$ (Stable)')
plt.plot(n_axis, x_ringing, 'g-s', label=r'$\Delta T = 1.9\tau$ (Ringing)')
plt.plot(n_axis, x_unstable, 'r-^', label=r'$\Delta T = 2.1\tau$ (Unstable)')

plt.axhline(1.0, color='k', linestyle='--', label='Theoretical Steady State')
plt.ylim(-1, 3)
```

Simulation Results

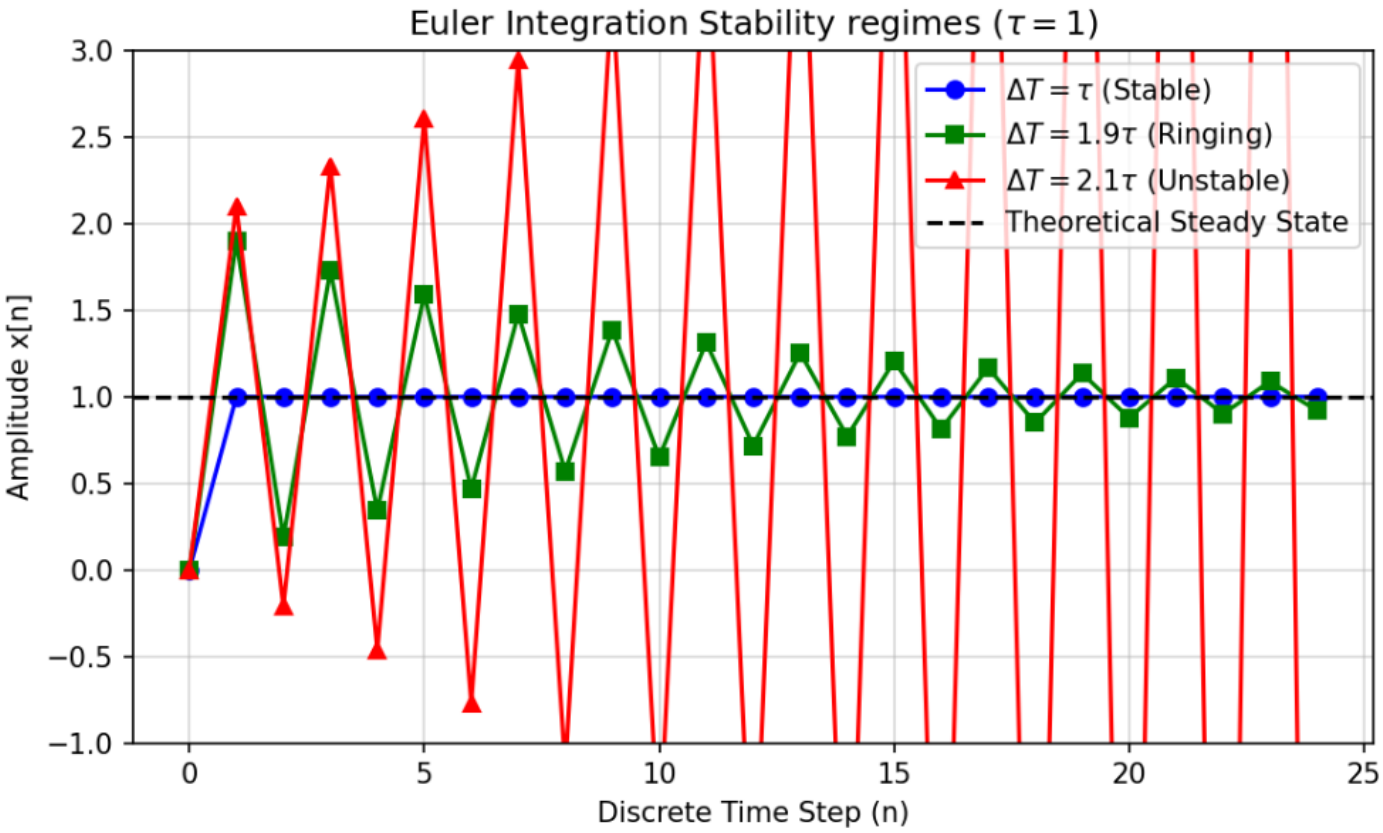

Figure: Computational proof of the 2τ boundary. Notice the severe oscillation (ringing) as ΔT approaches 2τ, and immediate divergence when it exceeds it.